

Data Mining Assignment III

Group 2

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Question 1 (5.2 in the book) Problem Statement

We will now derive the probability that a given observation is part of a bootstrap sample. Suppose that we obtain a bootstrap sample from a set of n observations.

(a)

What is the probability that the first bootstrap observation is not the j th observation from the original sample? Justify your answer.

Each draw is independent and uniform over the n observations, so the probability the first bootstrap draw equals observation j is $1/n$. Hence

$$P(\text{1st draw} \neq j) = 1 - \frac{1}{n}.$$

(b)

What is the probability that the second bootstrap observation is not the j th observation from the original sample?

Sampling is with replacement, so every draw is an independent, identical draw from the full set of n observations. The second draw has the same distribution as the first, so

$$P(\text{2nd draw} \neq j) = 1 - \frac{1}{n}.$$

(c)

Argue that the probability that the j th observation is not in the bootstrap sample is $(1 - 1/n)^n$.

The bootstrap sample consists of n independent draws. Observation j is *not* in the sample only if it is missed on every one of the n draws. By independence, the probabilities multiply:

$$P(j \notin \text{sample}) = \left(1 - \frac{1}{n}\right)^n.$$

(d)

When $n = 5$, what is the probability that the j th observation is in the bootstrap sample?

$$1 - \left(1 - \frac{1}{5}\right)^5 = 1 - (0.8)^5 = 1 - 0.32768 \approx 0.6723.$$

(e)

When $n = 100$, what is the probability that the j th observation is in the bootstrap sample?

$$1 - \left(1 - \frac{1}{100}\right)^{100} = 1 - (0.99)^{100} \approx 1 - 0.3660 = 0.6340.$$

(f)

When $n = 10,000$, what is the probability that the j th observation is in the bootstrap sample?

$$1 - \left(1 - \frac{1}{10000}\right)^{10000} \approx 1 - 0.36788 = 0.63212.$$

As $n \rightarrow \infty$, $\left(1 - \frac{1}{n}\right)^n \rightarrow e^{-1}$, so the probability converges to

$$1 - e^{-1} \approx 0.6321.$$

(g)

Create a plot that displays, for each integer value of n from 1 to 100,000, the probability that the j th observation is in the bootstrap sample. Comment on what you observe.

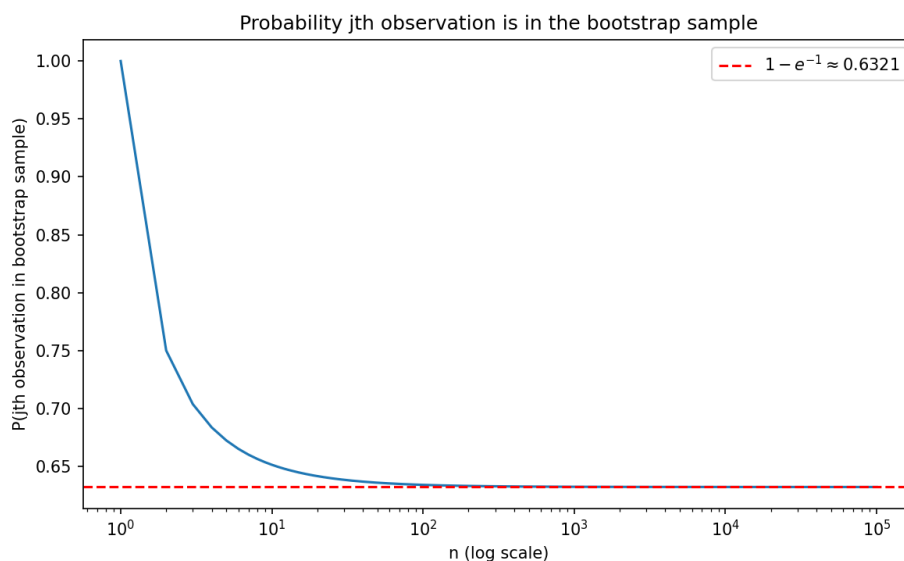


Figure 1: Probability that the j th observation appears in a bootstrap sample, for $n = 1$ to 100,000 (log scale on n).

Comment: The probability falls sharply for small n and then flattens out almost immediately, converging to the asymptote $1 - e^{-1} \approx 0.6321$. By the time n is in the hundreds, the curve is already essentially at its limiting value, and it barely changes even as n grows to 100,000. This confirms that the $\approx 63.2\%$ inclusion probability is a stable property of bootstrapping rather than something that depends strongly on dataset size.

(h)

```
rng = np.random.default_rng(10)
store = np.empty(10000)
for i in range(10000):
    store[i] = np.sum(rng.choice(100, size=100, replace=True) == 4) > 0
np.mean(store)
```

Running this gives

```
np.mean(store) ≈ 0.6362,
```

which matches the theoretical value from part (e), 0.6340, closely.

Comment: Note that the code as originally printed in the exercise, `rng.choice(100, replace=True)`, omits the `size=100` argument and therefore draws only a *single* value per iteration rather than a full bootstrap sample of size 100. That version instead estimates $P(\text{single draw} = 4) \approx 1/100 = 0.01$, not the probability that observation 4 is in a full bootstrap sample. With the corrected line above (size 100 draws per iteration), the simulation reproduces the theoretical answer from part (e) almost exactly, numerically confirming that roughly 63.6% of the 10,000 simulated bootstrap samples contained observation #4.

Question 2.(5.7 in the book) Problem Statement

In Sections 5.1.2 and 5.1.3, we saw that the `cross_validate()` function can be used to compute the LOOCV test error estimate. Alternatively, one could compute those quantities using just `sm.GLM()` and the `predict()` method of the fitted model within a for loop. We take this approach to compute the LOOCV error for a simple logistic regression model on the `Weekly` data set. Recall that in the context of classification problems, the LOOCV error is given by

$$CV_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{Err}_i.$$

(a)

*Fit a logistic regression model that predicts **Direction** using **Lag1** and **Lag2**.*

Fitting the model on all $n = 1089$ observations gives the coefficients:

$$\hat{\beta}_0 \approx 0.2212, \quad \hat{\beta}_{\text{Lag1}} \approx -0.0387, \quad \hat{\beta}_{\text{Lag2}} \approx 0.0602.$$

(b)

*Fit a logistic regression model that predicts **Direction** using **Lag1** and **Lag2** using all but the first observation.*

Refitting with observation 1 removed gives coefficients that are very close to the full-data fit:

$$\hat{\beta}_0 \approx 0.2232, \quad \hat{\beta}_{\text{Lag1}} \approx -0.0384, \quad \hat{\beta}_{\text{Lag2}} \approx 0.0608.$$

(c)

Use the model from (b) to predict the direction of the first observation. You can do this by predicting that the first observation will go up if $P(\text{Direction} = \text{"Up"} \mid \text{Lag1}, \text{Lag2}) > 0.5$. Was this observation correctly classified?

The predicted probability is $P(\text{Up}) \approx 0.5714$, so the model predicts **Up**. The actual observed direction for the first observation is **Down**. Hence this observation was **incorrectly classified**.

(d)

Write a for loop from $i = 1$ to $i = n$, where n is the number of observations in the data set, that performs each of the following steps:

- i. Fit a logistic regression model using all but the i th observation to predict *Direction* using *Lag1* and *Lag2*.
- ii. Compute the posterior probability of the market moving up for the i th observation.
- iii. Use the posterior probability for the i th observation to predict whether or not the market moves up.
- iv. Determine whether or not an error was made in predicting the direction for the i th observation. If an error was made, indicate this as a 1, and otherwise indicate it as a 0.

For each of the $n = 1089$ observations, we refit the logistic regression on the remaining $n - 1$ observations, predict the held-out observation's direction using the 0.5 probability threshold, and record a 1 if the prediction was wrong and a 0 if it was correct. (See accompanying notebook for the implementation.)

(e)

Take the average of the n numbers obtained in (d)iv in order to obtain the LOOCV estimate for the test error. Comment on the results.

Averaging the n error indicators gives the LOOCV test error estimate:

$$CV_{(n)} \approx 0.4500.$$

Comment: The LOOCV estimate of the test error rate is approximately 45%, meaning the logistic regression model using only *Lag1* and *Lag2* misclassifies the direction of the market on roughly 45% of the 1,089 weekly observations when each is left out and predicted using a model trained on the rest. This is only marginally better than the 50% error rate we would expect from random guessing, indicating that *Lag1* and *Lag2* alone have very little real predictive power for weekly market direction. This is consistent with the well-known difficulty of predicting stock market movements from a small number of lagged return values.